



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-21

Experiment 21. Bird Can Fly or Not

Aim

To write a Prolog program to determine whether a particular bird can fly or not.

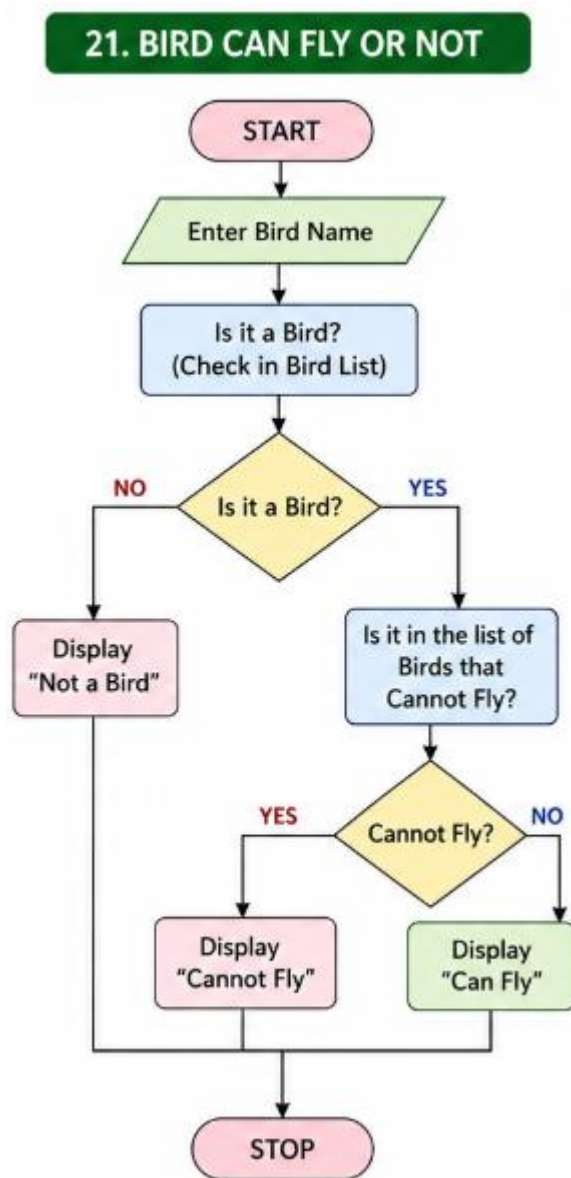
Objectives

- To understand Prolog facts and rules.
- To classify birds based on their flying ability.
- To use logical reasoning in Prolog.

Algorithm

1. Define different birds.
2. Define birds that cannot fly.
3. Check whether the given bird is a bird.
4. Check whether it belongs to the non-flying list.
5. Display whether it can fly.

Flowchart



Code

bird(parrot).

bird(eagle).

bird(pigeon).

bird(penguin).

bird(ostrich).

cannot_fly(penguin).

cannot_fly(ostrich).

can_fly(X) :-

bird(X),

\+ cannot_fly(X).

main :-

Bird = parrot,

can_fly(Bird),

write(Bird),

write(' can fly.').

:- initialization(main).

Output



The screenshot shows the OneCompiler Prolog IDE. The main editor displays a Prolog program in a file named 'HelloWorld.pl'. The program defines a list of birds, a predicate for birds that cannot fly, and a main predicate that checks if a specific bird can fly. The output pane on the right shows the result of the query: 'parrot can fly.'

```
1 bird(parrot).
2 bird(eagle).
3 bird(pigeon).
4 bird(penguin).
5 bird(ostrich).
6
7 cannot_fly(penguin).
8 cannot_fly(ostrich).
9
10 can_fly(X) :-
11     bird(X),
12     \+ cannot_fly(X).
13
14 main :-
15     Bird = parrot,
16     can_fly(Bird),
17     write(Bird),
18     write(' can fly.').
19
20 :- initialization(main).
```

Output:
parrot can fly.

Result

parrot can fly.

Conclusion

Thus, the Prolog program successfully determines whether a bird can fly or not.